

EXP 21 – Towers of Hanoi (Prolog)

Name: Hemalakshmi(192571049)

Aim

To write a Prolog program that implements the Towers of Hanoi algorithm using recursion and displays the sequence of moves required to transfer N disks from the source peg to the destination peg following the rules of the puzzle.

Algorithm

1. Start with three pegs: From, To, and Via.
2. If $N = 0$, stop (no moves required).
3. Recursively move $N-1$ disks from From to Via using To as auxiliary.
4. Move the largest disk (disk N) from From to To.
5. Recursively move the $N-1$ disks from Via to To using From as auxiliary.
6. Continue until all disks are moved to the destination peg.

Source Code

```
prolog
```

```
hanoi(0, _, _, _).
```

```
hanoi(N, From, To, Via) :-
```

```
    N > 0,
```

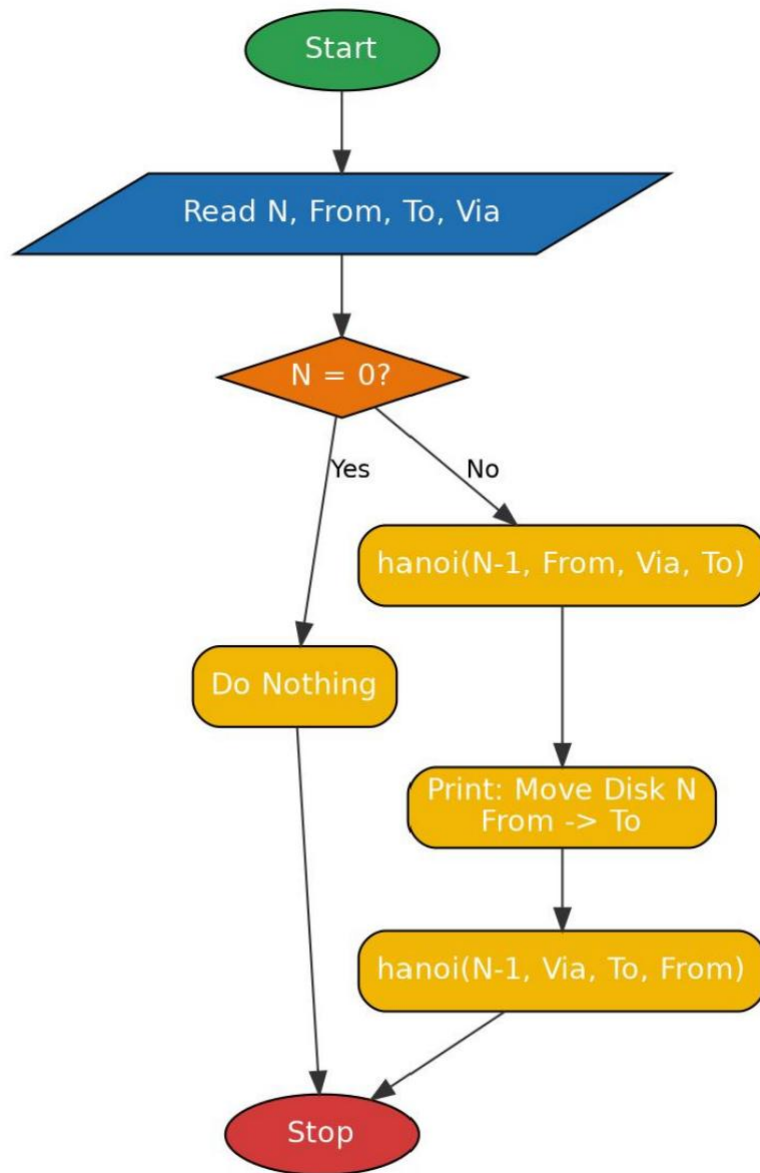
```
    N1 is N - 1,
```

```
    hanoi(N1, From, Via, To),
```

```
    format("Move disk ~w from ~w to ~w~n", [N, From, To]),
```

```
    hanoi(N1, Via, To, From).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [hanoi].
true.

?- hanoi(3, left, right, middle).
Move disk 1 from left to right
Move disk 2 from left to middle
Move disk 1 from right to middle
Move disk 3 from left to right
Move disk 1 from middle to left
Move disk 2 from middle to right
Move disk 1 from left to right
true.

?- hanoi(4, left, right, middle).
Move disk 1 from left to middle
Move disk 2 from left to right
Move disk 1 from middle to right
Move disk 3 from left to middle
Move disk 1 from right to left
Move disk 2 from right to middle
Move disk 1 from left to middle
Move disk 4 from left to right
Move disk 1 from middle to right
Move disk 2 from middle to left
Move disk 1 from right to left
Move disk 3 from middle to right
Move disk 1 from left to middle
Move disk 2 from left to right
Move disk 1 from middle to right
true.
```

Result:

The program successfully prints the exact sequence of moves required to shift all disks from the source peg to the destination peg using the auxiliary peg. It demonstrates how recursion breaks the problem into smaller moves and completes the full Towers of Hanoi solution.